Runbook and Script Reference

Project 3: PowerShell Automation for Active Directory

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Purpose: a reusable guide for running the toolkit: what to check first, the standard procedure, the commands for each script, how to read the reports and what to do when something goes wrong. This is the template-style document of the set. The screenshots that prove each script works are in Document 2 (JML Operations Log) and Document 3 (Audit and Validation Report).

1. Before you start

- Work on the domain controller (or a machine that has the ActiveDirectory PowerShell module) and open Windows PowerShell **as Administrator**.
- Use an account that is allowed to create, change and move users and to edit group membership.
- Make sure Config.ps1 matches the domain. Run the environment check first. Every department should show OU path OK and Group OK:

```
cd C:\PowerShell-AD-Automation\Scripts
.\Test-Environment.ps1
```

- Input CSV files live in the CSV folder. Logs and reports are written to the Logs folder.

2. Folder layout

```
PowerShell-AD-Automation\
  Scripts\   Config.ps1, Logging-Module.psm1, Shared-Functions.psm1,
             Test-Environment.ps1, New-BulkUsers.ps1,
Invoke-Mover.ps1,
             Invoke-Offboarding.ps1, Get-AuditReports.ps1
  CSV\       NewHires.csv, Movers.csv, Leavers.csv
             (plus three _Messy.csv files used for testing)
  Logs\      created at run time: logs, reports, hand-off file
```

3. Input files

Save each file as a plain CSV with a header row. Department names must match the ones in Config.ps1 (IT, Finance, HR, Management). Capital letters do not matter.

Joiner: NewHires.csv (all four columns are required)

```
FirstName,LastName,Department,JobTitle
Aarav,Mehta,IT,Support Technician
```

Mover: Movers.csv (Username and NewDepartment are required; the other two are optional)

```
Username,NewDepartment,NewJobTitle,TicketRef
amehta,Management,IT Operations Lead,TKT-2001
```

Leaver: Leavers.csv (only Username is required; Reason and TicketRef are optional but recommended)

```
Username,Reason,TicketRef
njoshi,Resignation,TKT-1001
```

Names may only contain letters, spaces, apostrophes, dots and hyphens. No digits.

4. Standard procedure (for every operation)

1. Open PowerShell as Administrator and go to the Scripts folder.
2. If the config or the domain has changed, run Test-Environment.ps1 first.
3. Check the CSV: right columns, no obvious typos, no duplicate people.
4. **Dry run** with -WhatIf. Read every line, and read the Note column in the table.
5. Run for real. For offboarding, read each Yes/No prompt before answering.
6. Verify the result in PowerShell and in Active Directory Users and Computers.
7. Keep the log and the report from the Logs folder. Hand over the temporary passwords and then do not keep or share the hand-off file.

5. Running each script

Joiner: onboarding new users

```
.\New-BulkUsers.ps1 -WhatIf
.\New-BulkUsers.ps1
.\New-BulkUsers.ps1 -CsvPath ..\CSV\OtherFile.csv
```

What good looks like: one green SUCCESS line for each account created and one for each group added, and a summary such as Created: 6 | Skipped: 0 | Failed/errors: 0. The temporary passwords are saved in a separate HANDOFF file in the Logs folder.

Checklist: accounts sit in the right OU, are in the right group, and are forced to change their password at first sign-in.

Mover: changing a user's department

```
.\Invoke-Mover.ps1 -WhatIf
.\Invoke-Mover.ps1
.\Invoke-Mover.ps1 -Username amehta -NewDepartment Management
-NewJobTitle "IT Operations Lead"
```

What good looks like: the new group added, the old department group removed, the department and title updated, the account moved to the new OU, and a summary such as Moved: 2. Any non-department groups are kept and logged as needing manager review. Read that warning and follow it up.

Checklist: the user's groups match the new department only, the title is correct and the account is in the right OU.

Leaver: offboarding users

```
.\Invoke-Offboarding.ps1 -WhatIf
.\Invoke-Offboarding.ps1
.\Invoke-Offboarding.ps1 -Username njoshi -Reason "Resignation"
```

The script asks "Are you sure you want to perform this action?" for each user. Y means yes for this user, A means yes to all and N means no for this user. Read the user's name in the prompt before answering.

What good looks like: for each user the steps appear in order (account disabled, password reset, groups removed, Description set, moved to Disabled Users) and the summary reads Offboarded with the number of users.

Checklist: the account is disabled, MemberOf is empty, the Description holds the date, reason and ticket, and the account is in the Disabled Users OU.

Audit: access review reports

```
.\Get-AuditReports.ps1
.\Get-AuditReports.ps1 -InactiveDays 7 -WarnDays 30
```

The script is read-only. It prints four reports and saves each one that has rows as a CSV in the Logs folder. Act on the findings separately: investigate each one and correct the access with the Mover or Leaver scripts, or by hand.

6. Script reference

Script	Purpose	Input	Key parameters	Output
Test-Environment.ps1	Checks the config against live Active Directory	None	None	Log
New-BulkUsers.ps1	Joiner: create users	NewHires.csv	-CsvPath, -WhatIf	Log, report CSV, hand-off CSV
Invoke-Mover.ps1	Mover: change department and access	Movers.csv	-Username, -NewDepartment, -NewJobTitle, -CsvPath, -WhatIf	Log, report CSV
Invoke-Offboarding.ps1	Leaver: disable and lock down	Leavers.csv	-Username, -Reason, -CsvPath, -WhatIf, -Confirm	Log, report CSV
Get-AuditReports.ps1	Four read-only audit reports	None	-InactiveDays (default 30), -WarnDays (default 14)	Log, up to four report CSVs

7. Reading the reports

Every row in a report has a Status and a Note that explains it.

Status	Meaning	What to do
Created, Moved, Offboarded	The operation succeeded	Verify in Active Directory
Created - group error	The account exists but was not added to a group	Add it to the group manually
DryRun, DryRun / Not confirmed	Nothing was changed (dry run, or the answer was No)	Review the plan, then run for real

Status	Meaning	What to do
Skipped - already exists, already offboarded, already in department	The work was already done	Nothing
Skipped - invalid row, invalid name	The row has missing or bad data	Fix the row in the CSV
Skipped - unknown department	The department is not in Config.ps1	Fix the spelling, or add it to Config.ps1
Skipped - config problem	A department's OU or group is missing in Active Directory	Run Test-Environment.ps1, then fix Config.ps1 or Active Directory
Skipped - duplicate row	The same person appears earlier in the file	Remove the duplicate row
Skipped - not found	The username does not exist	Check the spelling
Refused - protected account, Domain Admin, account disabled	The script will not touch these accounts	Handle them manually if it is genuinely needed
Failed	An Active Directory change failed. The Note lists the steps that finished	Follow the Note, finish the job manually and check the user's access

8. Security rules

- Never upload, email or screenshot the HANDOFF file. It holds temporary passwords.
- Never skip the dry run.
- Read every confirmation prompt before answering it.
- Do not offboard or move Domain Admins with these scripts. Handle privileged accounts by hand.
- Disable accounts and keep them. Do not delete them during offboarding.
- Run PowerShell with only the rights the job needs.
- Keep the logs and reports. They are the audit trail.

- Hand temporary passwords over securely, and make sure users change them at first sign-in.

9. Troubleshooting

What you see	Likely cause	Fix
The command was not found, but does exist in the current location	You left out .\ in front of the script name	Type .\New-BulkUsers.ps1
The term is not recognised, with a script name in red	A typo in the script name	Press Tab to complete the name
Running scripts is disabled on this system	PowerShell's execution policy	Run Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass (this window only)
CSV file not found, or missing required column(s)	Wrong path, or the header row is wrong	Check the path or -CsvPath, and compare the headers with section 3
Disabled Users OU not found	The OU is missing or the path in Config.ps1 is wrong	Fix Config.ps1 or create the OU. The script stopped before changing anything
Skipped - config problem	A department's OU or group does not exist	Run Test-Environment.ps1 and fix the name
Access denied	The account does not have the rights	Run PowerShell as an account that is allowed to manage users
Audit shows many Never logged on	The accounts have not signed in, or the last logon date has not refreshed yet (it updates about every 14 days)	Normal in a lab. Check again later

10. Glossary

Term	Meaning
JML	Joiner-Mover-Leaver: the stages of an employee's account lifecycle
OU	Organizational Unit: a folder in Active Directory
Security group	A set of users that permissions are given to
-WhatIf	PowerShell's dry-run switch: shows what would happen without doing it
Privilege creep	Old access that is never removed after a role change
Least privilege	Users have only the access their current role needs
HANDOFF file	The separate CSV that holds temporary passwords. It is never uploaded
ADUC	Active Directory Users and Computers, the graphical admin tool